

EDA and Feature Engineering Report

Machine Learning-Based Network Intrusion Detection

This report documents the exploratory data analysis, data-quality assessment, feature engineering, feature selection and dimensionality-reduction work performed for the UNSW-NB15 network intrusion detection project. It is designed as a standalone public-facing analytical reference.

Purpose

Formal reference document for the public project repository.

1. Objective and Dataset

The objective of this stage is to understand the network traffic data, identify patterns relevant to binary intrusion detection, prepare variables for modelling, select informative features and reduce dimensionality where appropriate.

The project uses the UNSW-NB15 dataset. The documented dataset contains 257,673 observations and 45 columns, with 175,341 observations in the training set and 82,332 in the testing set.

2. Data Quality Assessment

The analysis reports no missing values in the training data and no exact duplicate rows in the original training dataset. Numerical variables have heterogeneous scales and distributions, including skewness and extreme values. Categorical variables require encoding before modelling.

The identifier `id` and the analysis variable `attack_cat` are kept outside the predictive feature set. The binary target is `label`, where 0 represents normal traffic and 1 represents malicious traffic.

3. Applied Exploratory Data Analysis

EDA examines target distribution, numerical feature distributions, categorical traffic characteristics, connection state and correlations. These observations support subsequent preprocessing and feature-engineering choices.

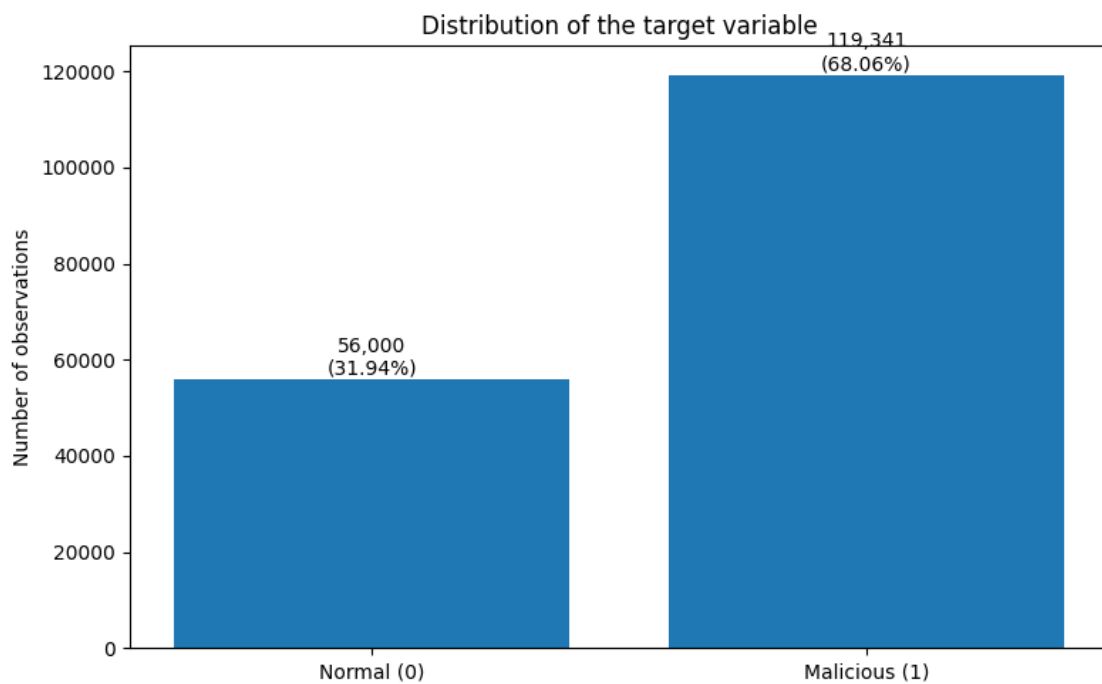


Figure 1. Target distribution in the training data.

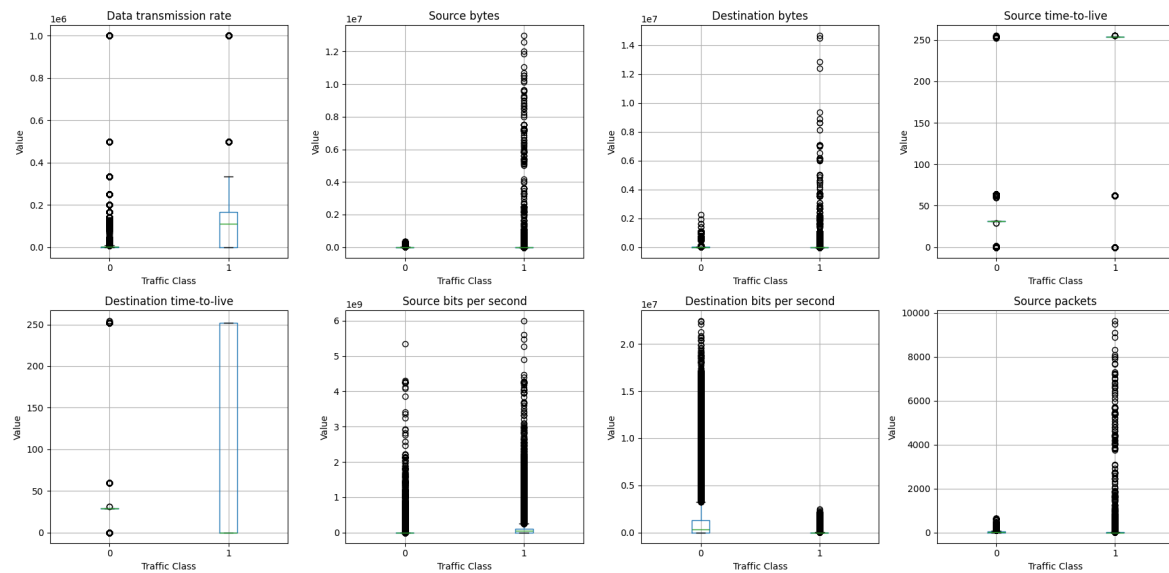


Figure 2. Numerical feature distributions examined during EDA.

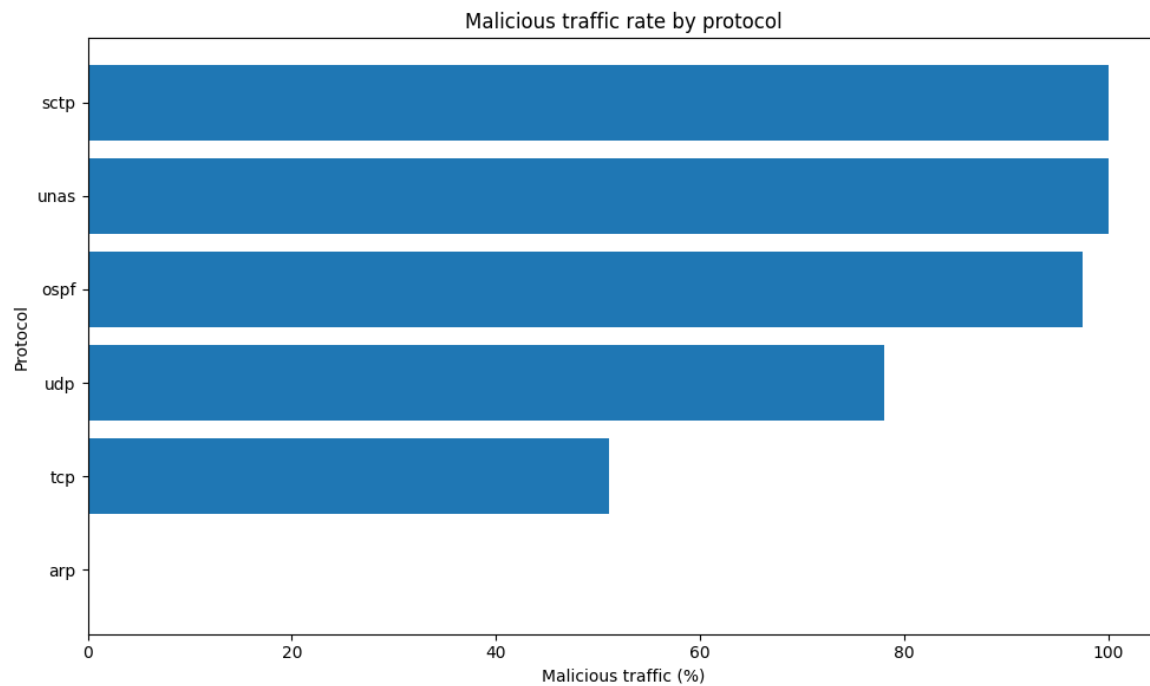


Figure 3. Protocol-level traffic characteristics.

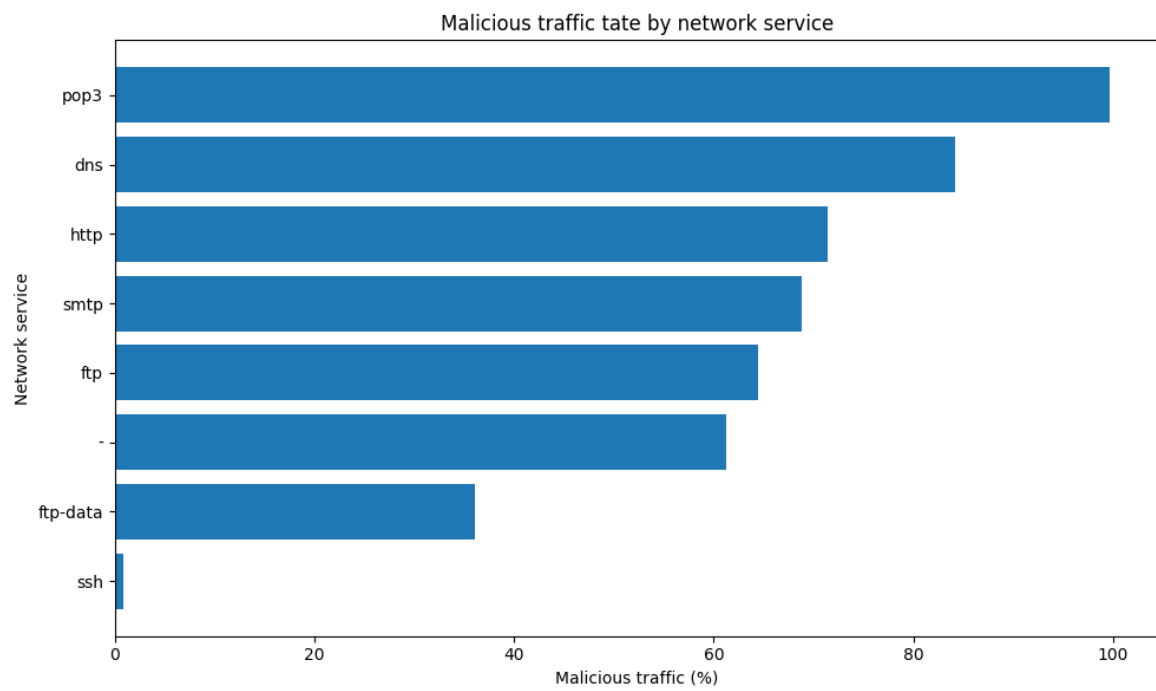


Figure 4. Service-level traffic characteristics.

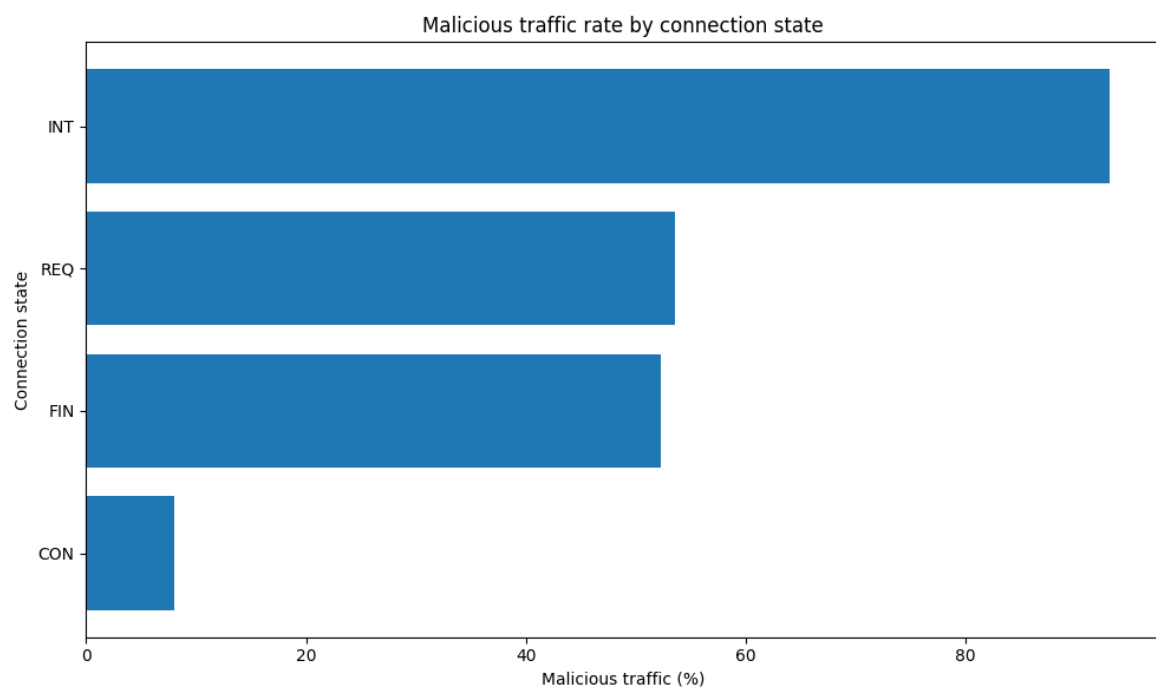
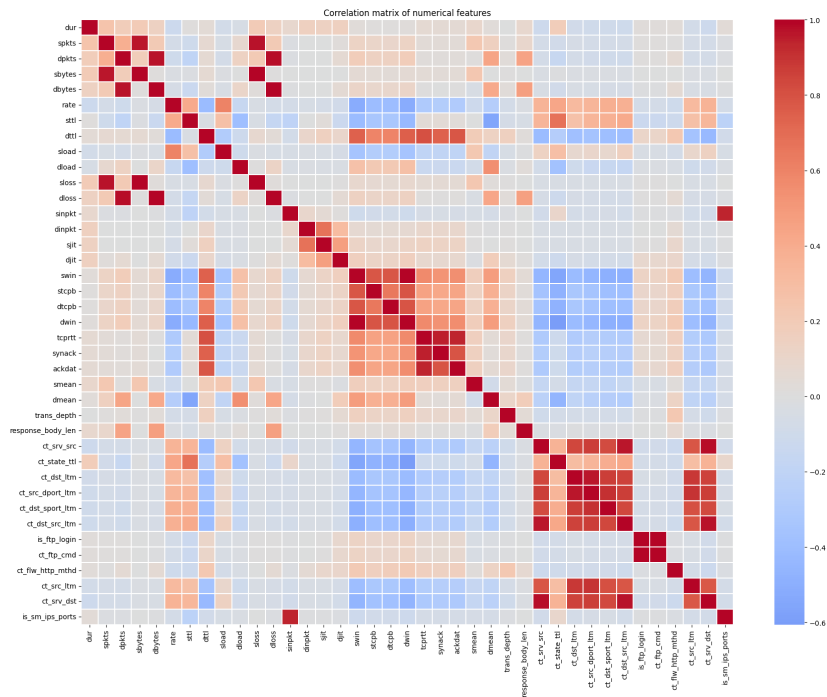


Figure 5. Connection-state characteristics.



4. Feature Engineering

Three network-derived features are documented in the modelling workflow: `total_bytes`, `total_packets` and `bytes_ratio`. These features combine or relate source and destination traffic measures to provide additional representations of the network flow.

Feature	Purpose
total_bytes	Combines source and destination bytes.
total_packets	Combines source and destination packet counts.
bytes_ratio	Captures the relative balance between the two directions of traffic.

5. Feature Selection

Mutual Information followed by SelectKBest is used as the documented feature-selection approach. The workflow retains the 30 highest-scoring features for subsequent analysis and modelling.

6. Dimensionality Reduction with PCA

PCA is applied after feature selection and scaling. The documented workflow retains 15 principal components, corresponding to 95.18% cumulative explained variance.

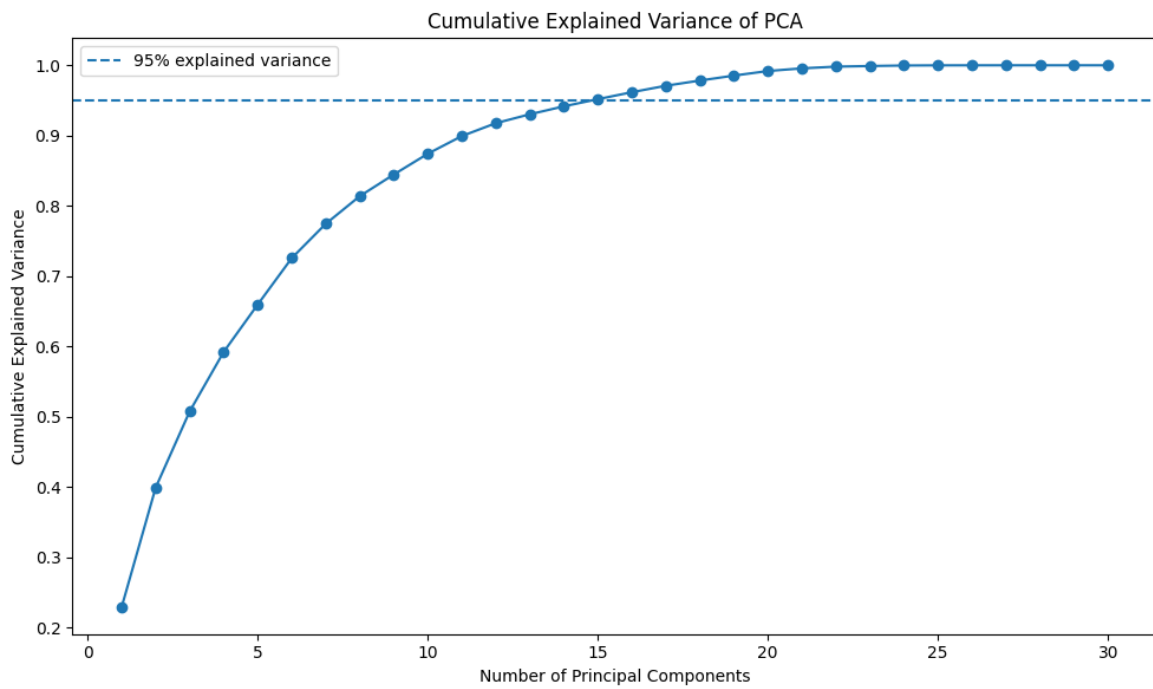


Figure 7. PCA explained variance and the retained component configuration.

7. Feature Importance and Explainability

Feature importance and explainability are addressed using the final Decision Tree model. Model-based feature importance identifies variables contributing most to model decisions, while SHAP provides global and local explanations of feature contributions.

Feature	Impurity-based Importance
sttl	88.26%
ct_srv_dst	5.93%
bytes_ratio	5.38%
Other features	0.43% combined

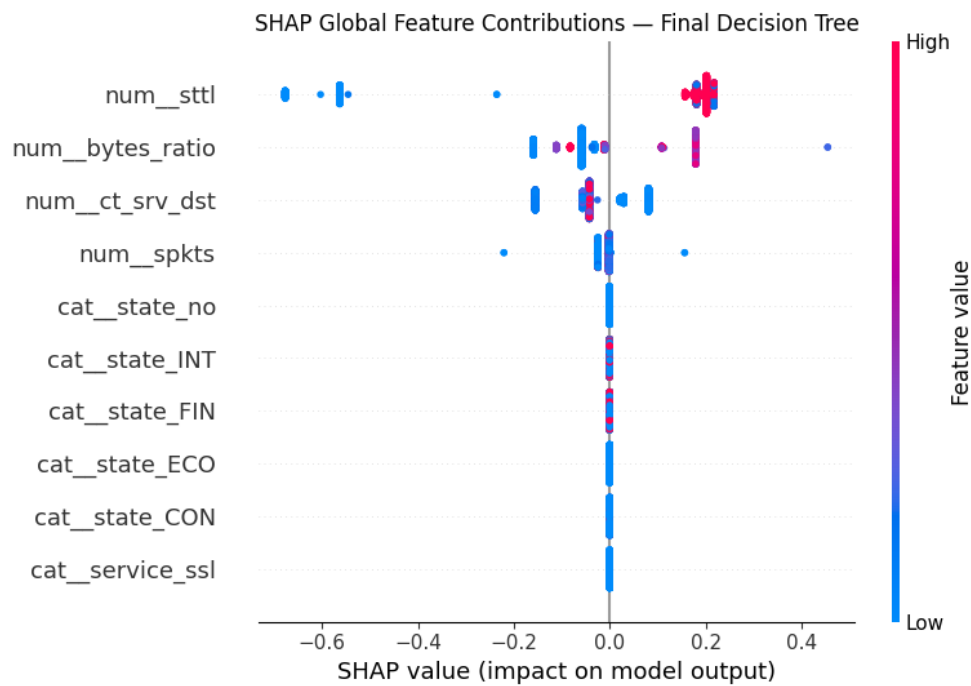


Figure 8. Global SHAP explanation of the final Decision Tree.

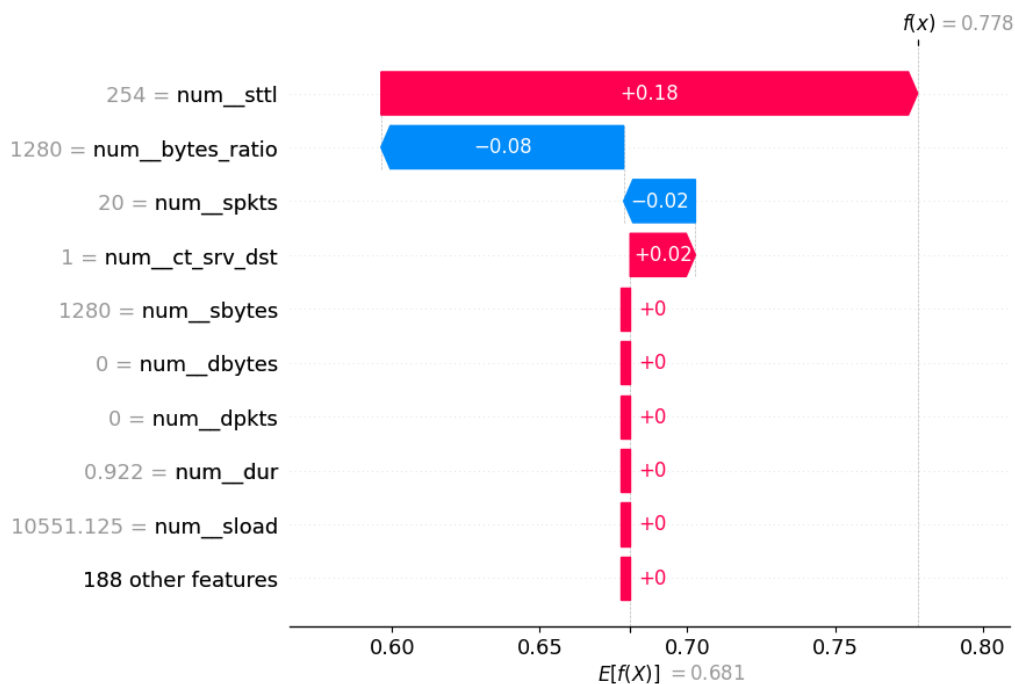


Figure 9. Example local SHAP explanation.

8. Key Findings and Justifications

The EDA identified heterogeneous numerical scales and distributions, categorical variables requiring encoding, and traffic characteristics that differ across normal and malicious observations. Feature engineering adds network-derived representations, while Mutual Information and SelectKBest reduce the feature space to 30 selected variables. PCA then represents the selected feature space using 15 components while retaining 95.18% of cumulative explained variance.

The final Decision Tree places strong importance on `st t l`. This concentration supports interpretability but also creates a robustness consideration: if the relationship learned from the UNSW-NB15 environment changes in another network environment, model behaviour may be affected.

9. Reproducibility and GenAI-Assisted Documentation

The analytical workflow is documented in the project notebook. This formal report is a documentation artefact derived from that existing analysis and its recorded results.

Stage	How GenAI was used	Control
Source review	Reviewed the documented EDA, feature engineering, feature selection, PCA and explainability content.	Only documented project content was retained.
Information extraction	Reorganised existing analytical information into a standalone report structure.	Numerical results and terminology were preserved.
Visual assembly	Selected figures already produced by the project analysis and integrated them into the report.	No new experimental results were generated for the report.
Editorial and layout	Improved hierarchy, captions, tables, pagination and visual consistency.	Scientific scope was not expanded.
Human validation	Checked content, terminology, numerical results and scope before publication.	Final publication remains subject to human review.

GenAI boundary. GenAI supported documentation, organisation, visual assembly and format transformation. It was not used to invent dataset results, model results or scientific conclusions.